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Minority stress among LGB people: Investigating relations among distal and proximal stressors

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Abstract
Research among lesbian, gay, or bisexual (LGB) individuals often distinguishes between distal (e.g., discrimination) and proximal (e.g., expectations of rejection, internalized homophobia, and concealment) stressors. There are few longitudinal studies, however, testing the causal distal-proximal link. The present study examined these proposed associations among a sample of 192 LGB adults. Cross-lagged structural equation modeling revealed that each stressor was stable over 1 month. Also, Time 1 discrimination negatively predicted Time 2 concealment, but Time 1 expectations of rejection positively predicted Time 2 discrimination and concealment. Time 1 concealment positively predicted Time 2 internalized homophobia. Our results suggest the need for further evaluation of the distal-proximal distinction. Practical implications and directions for future research are discussed.

Keywords Discrimination · LGB · Minority stress

People in the United States (U.S.) have reported more support for lesbian, gay, bisexual, or transgender (LGBT) people, but about 37% of Americans still report anti-LGBT prejudice (Brown 2017). After the U.S. presidential election in 2016, LGBT people reported reductions in thriving from 51% to 41% (Gates 2016). Minority stress frameworks (Hatzenbuehler 2009; Meyer 2003) outline how such widespread prejudice against sexual minorities is associated with increased risks for various mental health outcomes (e.g., Brewster et al. 2013; Douglass et al. 2017; Velez and Moradi 2016), but research examining associations among the stressors themselves is limited. This study examines how minority stressors are associated over time.

Data and ideas presented in this manuscript have not been previously disseminated

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Theoretical Frameworks

The Minority Stress Model (Meyer 2003) describes stress processes among LGB people that contribute to a higher prevalence of mental health concerns compared to heterosexual people. LGB people encounter distal and proximal *minority* stress, which extends beyond the general stress that everyone experiences. Distal stressors are external prejudice events directed toward a person (e.g., heterosexist discrimination, victimization, and homonegative microaggressions). Proximal stressors are a person's internal or subjective responses (Meyer 2003). Distal stressors refer to an event (e.g., being called a heterosexist slur), and proximal stressors involve an appraisal of the event (e.g., internalizing LGB bias). There are three major proximal stressors: expectations of rejection, internalized homophobia, and concealment of one's LGB identity. Expectations of rejection is the extent to which people expect to be stigmatized by others, internalized homophobia regards self-directed antigay attitudes, and concealment is hiding an LGB identity (Meyer 2003). Distal stressors are theorized to contribute to the development of proximal stressors.

The Psychological Mediation Framework (Hatzenbuehler 2009) also positions distal stressors as *predictors* of proximal stressors, which in turn predict greater risk of negative mental health outcomes. The positioning of proximal stressors as mediators between distal stressors and mental health outcomes implies causality (Meyer 2003), but relatively little attention

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has been devoted to testing this proposition through longitudinal research. Distal and proximal stressors may share bidirectional associations (Hatzenbuehler 2009). Discrimination, for example, may be influenced by contextual, personal, and structural factors, suggesting that variables such as expectations of rejection may predict discrimination (Major and O'Brien 2005). Thus, longitudinal data are needed to examine such causal models as cross-sectional data are unable to establish patterns of causality.

Distal Versus Proximal Stressors

Cross-sectional research using minority stress frameworks has positioned proximal stressors as mediators of the links between heterosexist discrimination and mental health outcomes suggesting that discrimination *precedes* the development of proximal stressors (Brewster et al. 2013; Denton et al. 2014; Douglass et al. 2019; Feinstein et al. 2012). Other research has suggested proximal-distal pathways (e.g., Ragins et al. 2007; Swank et al. 2013) or has positioned distal and proximal stressors on the same level of analysis (e.g., Velez et al. 2013). One study found that a model with distal and proximal stressors on the same level, compared with a mediation model where distal stressors preceded proximal stressors, both had comparable fit indices and no significant differences in fit based on chi-square nested comparisons alone (Mereish and Poteat 2015). Still, determining causality based on any of these cross-sectional studies is difficult.

Some longitudinal studies grounded in minority stress frameworks have examined how distal or proximal stressors are associated with physical or mental health outcomes (e.g., Dyar et al. 2019; Mohr and Sarno 2016; Pachankis et al. 2018; Puckett et al. 2016). These studies are usually focused on how minority stressors are associated with various outcomes and have helped demonstrate the negative effects of sustained minority stressors. Less emphasized, however, are the associations between minority stressors. Focusing on how these stressors influence one another can help to substantiate or modify existing theory, as the notion of reciprocal associations among minority stressors (e.g., Hatzenbuehler 2009; Major and O'Brien 2005) has not been comprehensively tested.

The Present Study

This study examines the distal-proximal stressor distinction proposed in the Minority Stress Model (Meyer 2003) and Psychological Mediation Framework (Hatzenbuehler 2009). We hypothesize that Time 1 discrimination will significantly predict greater levels of Time 2 expectations of rejection, internalized homophobia, and concealment of LGB identity. There are also two exploratory aims: (1) examine potential bidirectional associations among stressors and (2) explore the temporal stability of minority stressors. This study helps evaluate the directionality of associations between distal and proximal stressors. Cross-lagged structural equation modeling will be used to assess the hypothesized links over 1 month. We chose a period of 1 month to extend on studies that have examined some minority stressors over the course of seven to 10 days (Mohr and Sarno 2016; Swim et al. 2009). A time frame of 1 month will be less sensitive to extreme fluctuations over the course of a week due to a single event while also capturing stressful experiences that could dissipate over a longer time period (Herek et al. 1999; Swim et al. 2009).

Method

Instruments

Discrimination Experiences of discrimination were measured with the 9-item Everyday Discrimination Scale (EDS; Williams et al. 1997), which uses a 6-point Likert scale ranging from *never* to *almost every day*. The original EDS specifically evaluated race and ethnicity; thus, we edited the scale instructions for the current study to read "In your day-to-day life, how often do any of the following things happen to you as a result of your sexual orientation?" Participants were asked to consider the events of the past month. Example items include, "You are threatened or harassed," and "You are called names or insulted." Williams et al. found an internal consistency reliability estimate of .88 and found evidence of validity as the scale correlated in the expected directions with stress and discrimination. This scale has been used in studies with sexual minority samples and has been found to be reliable and valid among these samples (e.g., Douglass et al. 2017; Frost et al. 2015). In the present study, the internal consistency reliability of scale scores was $\alpha = .93$ at Time 1 and $\alpha = .95$ at Time 2.

Expectations of Rejection The 10-item Stigma Consciousness Questionnaire (SCQ; Pinel 1999), which uses a 7-point Likert scale, was used to assess the degree to which participants expected rejection because of being LGB. Item choices ranged from *strongly disagree* to *strongly agree*, and sample items include, "I never worry that my behaviors will be viewed as stereotypically LGB," and "When interacting with people, I feel like they interpret all my behaviors in terms of the fact that I am a LGB person." The SCQ is often used to measure expectations of rejection among LGB people (e.g., Berghe et al. 2010; Douglass et al. 2017; Lewis et al. 2014). Some scholars have distinguished expectations of rejection from the similar concept of rejection sensitivity by arguing that expectations of rejection is a cognitive process whereas rejection sensitivity also encompasses affective responses (e.g., anxiety; Dyar et al. 2016). This study follows the

example of previous scholars who employed the SCQ to assess expectations of rejection.

In the original scale construction article for the SCQ, reliability of scale scores was determined with an internal consistency reliability estimation of .81. Validity in the scale's responses was also demonstrated through correlations in expected directions with measures of stigma and discrimination (Pinel 1999). In the present study, the internal consistency reliability of scale scores was $\alpha = .78$ at Time 1 and $\alpha = .81$ at Time 2.

Concealment The 10-item Self-Concealment Scale (SCS; Larson and Chastain 1990) was used to measure the degree to which participants hid being LGB from others. Item choices used a 5-point Likert scale, ranging from *strongly disagree* to *strongly agree*. We adapted items to be specific to concealment of LGB identity such as "Being LGB is so private I would lie if anybody asked me about it" instead of "I have a secret that is so private I would lie if anybody asked me about it." Larson and Chastain found a Cronbach's alpha of .83 and found evidence of test-retest reliability in the responses over 4-weeks ($r = .81$). Predictive validity was demonstrated through positive correlations with psychological distress and negative correlations with social support. Factor analyses provided evidence of unidimensionality and showed that responses on the SCS were different than responses on a separate measure of self-disclosure (Larson and Chastain 1990). In the present study, the internal consistency reliability of scale scores was $\alpha = .91$ at Time 1 and $\alpha = .92$ at Time 2.

Internalized Homophobia Self-stigma that was associated with participants' sexual orientation was measured using the 5-item Revised Internalized Homophobia Scale (IHP-R; Herek et al. 2009), which was developed from the original Internalized Homophobia Scale developed for use with gay men (Martin and Dean 1992). Scale items used a 5-point Likert scale, ranging from *strongly disagree* to *strongly agree*, and sample items include, "I wish I weren't gay/lesbian/bisexual," and "I feel that being gay/lesbian/bisexual is a personal shortcoming for me." Herek et al. (2009) found evidence of reliability in the internal consistency estimate of .82, responses were strongly correlated with responses to the original IHP, and test-retest reliability was found in follow-up surveys administered after 1 year ($r = .67$). Analyses of the IHP-R also showed a similar factor structure to the IHP, which has been found to have good internal consistency reliability and construct validity (Herek et al. 1998). In the present study, the internal consistency reliability of scale scores was $\alpha = .89$ at Time 1 and $\alpha = .88$ at Time 2.

Procedure

Methods and procedures were approved by an Institutional Review Board. We recruited participants using Amazon's online data collection service Mechanical Turk (MTurk), which is a useful method when sampling hard-to-reach populations, such as LGB people (Smith et al. 2015). MTurk participants have been found to be at least as reliable and diverse as traditional samples (e.g., community; Buhrmester et al. 2011; Goodman et al. 2013).

MTurk workers are compensated for completing Human Intelligence Tasks (HITs). We created a HIT for workers that were at least 18 years old, currently resided in the United States, identified as lesbian, gay, or bisexual, and were willing to participant in a follow-up survey. Workers could click the survey link and were directed to an informed consent page where they could choose to participate. Workers who completed our first survey were contacted 1 month later for the follow-up survey. Participants received $0.30 for each survey.

Of the 484 participants who completed our first survey 22 were removed for incorrectly responding to one of two attention checks (e.g., "Select agree if you are paying attention"). At the end of the survey we also asked participants if their responses should be excluded for any reason—four participants identified as heterosexual and were removed from the sample. After removals, 458 participants were invited to take the second survey and 196 people responded. Of these responses, four participants were removed for failing one of two attention checks resulting in a final sample of 192 people. Demographic information for these participants can be found in Table 1.

Results

Preliminary Analyses

Univariate outliers were cases with standardized scores outside |3.29|, and multivariate outliers were cases with a Mahalanobis distance significant at $p < .001$ at the corresponding χ^2 value (Tabachnick and Fidell 2013). Three univariate outliers were detected but were not values outside the range of possible scores. Because about 1% of extreme values are expected in larger sample sizes (Tabachnick and Fidell 2013; Osborne and Overbay 2004), and these three cases comprised about 0.2% (3/1536) of possible values, we retained these cases for analyses. Three cases also emerged as multivariate outliers at $p < .001$. Because conducting analyses with and without these multivariate outliers yielded no significant differences in model fit or significance, direction, or magnitude of associations we retained these cases for analyses.

We examined the normality of variables by examining skewness and kurtosis coefficients and normality plots (Weston and Gore 2006). No variables approached skewness > |3| or kurtosis values > |10| and appeared normally distributed (Weston and Gore 2006). Bivariate correlations among

Table 1 Demographic characteristics of participants

Characteristic	*N*	%
Participants	192	
Sexual Orientation		
Lesbian	43	22.4
Gay	57	29.7
Bisexual	92	47.9
Race/Ethnicity		
African American	21	10.9
Asian American	13	6.8
Asian Indian	3	1.6
Hispanic American	12	6.3
Native American	7	3.6
White	152	79.2
Gender		
Cisgender Man	84	43.8
Cisgender Woman	94	49
Genderqueer	8	4.2
Transgender	4	2.1
Other	2	1
Education		
Less than High School	1	0.5
Some High School	2	1
Completed High School	21	10.9
Trade or Vocational School	10	5.2
Some College	56	29.2
Completed College	67	34.9
Professional Degree	35	18.2
Social Class		
Lower Class	23	12
Working Class	71	37
Middle Class	82	42.7
Upper Middle Class	14	7.3
Upper Class	2	1
Relationship Status		
Single/Never Married	95	49.5
Married	48	25
Living with a Partner	41	21.4
Divorced	8	4.2

Some percentages may exceed 100% due to selecting multiple categories, and some may not sum to 100% due to missing data

all study variables at Time 1 and Time 2, along with means and standard deviations, are displayed in Table 2.

To examine if attrition was associated with age, education, gender, race, social class or the study variables, we used independent samples *t* tests and chi-squared tests to compare those who completed the survey at just Time 1 to those who completed both surveys. There were not enough participants in each race category to conduct comparisons that would be adequately powered across all groups, so we created a dummy variable with one group consisting of all non-White participants and another group with White participants. Independent samples *t* tests showed that those who dropped out were significantly younger ($M = 30.23$, $SD = 8.65$ vs. $M = 33.73$, $SD = 10.44$, $p < .001$), and had significantly higher mean scores of discrimination ($M = 32.76$, $SD = 15.80$ vs. $M = 26.48$, $SD = 11.09$ $p < .001$) and internalized homophobia ($M = 10.41$, $SD = 5.14$ vs. $M = 8.77$, $SD = 4.31$ $p < .001$). There were no significant differences regarding expectations of rejection ($p = .526$), concealment ($p = .138$), gender ($p = .203$), education ($p = .148$), and social class ($p = .752$), but more non-white participants dropped out of the study ($\chi^2(1) = 6.10$, $p = .013$). The significantly higher levels of discrimination and internalized homophobia are in line was past research suggesting higher levels of distress contributes to greater levels of attrition (Gustavson et al. 2012).

We also examined group differences between LGB people on the four minority stressors measured at Time 1. Bisexual participants reported significantly lower levels of expectations of rejection ($M = 36.97$, $SD = 9.26$) when compared with lesbian ($M = 43.07$, $SD = 7.72$, $p = .001$) and gay ($M = 42.44$, $SD = 9.72$, $p = .002$) participants. Bisexual participants reported significantly greater levels of concealment than lesbian participants ($M = 23.08$, $SD = 9.15$ vs. $M = 18.82$, $SD = 8.07$, $p = .037$) and significantly lower levels of discrimination than gay participants ($M = 16.07$, $SD = 7.94$ vs. $M = 20.40$, $SD = 8.32$, $p = .005$). There were no significant group differences on scores of internalized homophobia ($p = .754$). Given the significant differences related to age and race above, and these significant differences regarding specific sexual orientation identity, we conducted all analyses with age, race, and sexual orientation included as covariates.

Missing Data

At Time 1 one participant was missing two data points and 10 participants were missing one data point. At Time 2 nine participants were missing one data point. To see if missingness was systematically associated with age, gender, or any of the study variables, we created a dummy-coded variable to compare participants with and without missing data at each time point and conducted an independent samples *t* test (Tabachnick and Fidell 2013). The missing and non-missing groups were not significantly different on any of the variables at either time point ($p > .05$). To address these missing data we used full information maximum likelihood estimation (FIML), which uses all available data to generate estimates with added error to produce unbiased estimates. FIML is considered superior to other methods of handling missing data such as listwise deletion and mean substitution, which tend to produce biased estimates (Tabachnick and Fidell 2013).

Table 2 Bivariate statistics and correlations between variables at Time 1 and Time 2

Variable	1	2	3	4	5	6	7	8
1. Discrimination T1	–							
2. Discrimination T2	.76**	–						
3. Expectations T1	.51**	.50**	–					
4. Expectations T2	.32**	.46**	.76**	–				
5. Homophobia T1	.22*	.28**	.07	.04	–			
6. Homophobia T2	.16*	.26*	.10	.08	.81**	–		
7. Conceal T1	.03	.14	.08	.10	.57**	.54**	–	
8. Conceal T2	−.02	.13	.11	.18*	.45**	.62**	.81**	–
M	18.16	17.75	39.95	39.72	8.77	8.64	21.35	20.59
SD	8.27	8.59	9.48	9.91	4.31	4.21	8.93	8.95

Expectations expectations of rejection, *Homophobia* internalized homophobia, and *Conceal* sexual orientation identity concealment. Significant at $*p < .05$ and $**p < .001$ levels

Model Testing

We used SEM in AMOS (Arbuckle 2014) to evaluate our models. To minimize the likelihood of Type I and Type II error, we selected the following fit indices: chi-square (χ^2), the Comparative Fit Index (CFI), the Tucker-Lewis Index (TLI) and the Root Mean Square Error of Approximation (RMSEA; Hu and Bentler 1999). A significant χ^2 can indicate a poor fitting model, but this test is often significant in larger samples (Tabachnick and Fidell 2013). Criteria for the CFI, TLI, and RMSEA have ranged from less conservative (CFI & TLI ≥ .90; RMSEA ≤ .10) to more conservative (CFI & TLI ≥ .95; RMSEA ≤ .08; Hu and Bentler 1999; Weston and Gore 2006).

Using parcels instead of item-level data when conducting latent variable SEM results in better model fit, less biased estimates, increased reliability, and more precise parameter estimates (Bandalos 2002). The five items that measured internalized homophobia were used as the indicators for the latent construct at both time points. Discrimination, expectations of rejection, and concealment, however, were comprised of more than five items. To maximize equality of factor loadings, we created three parcels each to use as indicators for these latent variables (Little 2013). We created parcels by using exploratory factor analysis on each measure and assigned items to parcels in countervailing order according to the size of the factor loading to ensure they would have approximately equivalent factor loadings (Weston and Gore 2006). We used the same items at both time points to create parcels.

Cross-lagged models consist of two types of paths: autoregressive and cross-lagged. Autoregressive paths are the relation of a variable to itself at subsequent time points (e.g., path from discrimination at Time 1 to itself at Time 2) and represent the stability of a variable. Cross-lagged paths represent the effect of one variable to a different variable over time (e.g., the effect of Time 1 discrimination on Time 2 concealment). These paths are estimated while controlling for one another whereas a bivariate correlation fails to account for other influences. Cross-lagged panel models allowed us to garner a sense of how minority stressors were associated over time (Kline 2016).

Because we tested models with two time points it was necessary to control for correlations of the variables at Time 1, the covariances at Time 2, and the cross-wave error terms of the same observed indicators (Cole and Maxwell 2003). We also ensured that our model satisfied assumptions of configural and weak factorial invariance. Configural invariance constrains the constructs to be in the same configurations at each time point and was tested with our measurement model. To test weak factorial invariance we constrained factor loadings to be equivalent across time and compared fit indices to the configural model to determine if model fit declined. This examination of invariance is an essential pre-condition in longitudinal designs (Chan 1998; Widaman et al. 2010).

Measurement Model

The measurement model – which was analogous to the configural model – had good fit to the data, χ^2 (368) = 637.72, $p < .001$, CFI = .95, TLI = .93, and RMSEA = .06, 90% *CI* [.05, .07]. All latent variable indicators loaded onto their respective factors at .63 or higher. This model was compared with the weak factorial invariance model, which also had good fit to the data χ^2 (378) = 653.79, $p < .001$, CFI = .94, TLI = .93, and RMSEA = .06, 90% *CI* [.05, .07]. All indicators loaded onto their respective factors at .65 or higher. This model produced a non-significant change in fit χ^2 (10) = 16.07, $p = .098$. Thus, the factor structures of the study constructs were consistent across time.

Structural Models

Our first structural model was a baseline model where each Time 1 variable only predicted its associated Time 2 variable. This model had good fit to the data, χ^2 (390) = 675.87, $p < .001$, CFI = .94, TLI = .93, and RMSEA = .06, 90% *CI* [.05, .07]. Each Time 1 variable significantly predicted itself at Time 2 with standardized estimates as follows: discrimination ($\beta = .78$, $SE = .05$, $p < .001$), expectations of rejection ($\beta = .73$, $SE = .06$, $p < .001$), internalized homophobia ($\beta = .80$, $SE = .04$, $p < .001$), and concealment ($\beta = .80$, $SE = .05$, $p < .001$). Next, we tested a reciprocal model that included paths from the baseline model *along with* paths from each of the variables at Time 1 to the other variables at Time 2. This model also had good fit to the data χ^2 (378) = 653.29, $p < .001$, CFI = .94, TLI = .93, and RMSEA = .06, 90% *CI* [.05, .07] and was a significant improvement compared with the baseline model $\Delta\chi^2$ (12) = 22.58, $p = .032$. As seen in Fig. 1, Time 1 discrimination significantly and negatively predicted Time 2 concealment ($p = .026$). Time 1 expectations of rejection significantly and positively predicted Time 2 discrimination ($p = .034$) and concealment ($p = .033$). Lastly, Time 1 concealment significantly and positively predicted Time 2 internalized homophobia ($p = .048$). Because deleting non-significant paths would be based on the specific variation within our sample, we did not run a final trimmed model (Goodboy and Kline 2017). Factor loadings and covariances for this final model can be found in Tables 3 and 4.

Discussion

In the Minority Stress Model (Meyer 2003) and Psychological Mediation Framework (Hatzenbuehler 2009) distal stressors (e.g., heterosexist discrimination) are positioned as predictors of proximal stressors (e.g., expectations of rejection, internalized homophobia, and concealment of LGB identity). None of our hypotheses were supported as Time 1 discrimination was only associated with Time 2 sexual orientation concealment but in the direction opposite our hypothesis. Greater scores of Time 1 discrimination were associated with lower Time 2 concealment scores. Although contrary to the Minority Stress Model, this finding aligns with research that has found heterosexist discrimination to predict greater levels of disclosure (or less concealment; Ragins et al. 2007). Also, identity disclosure occurs along a continuum (e.g., partial versus full disclosure) and can have positive and negative outcomes (Ragins 2008). For people already experiencing heterosexist discrimination concealing one's LGB identity may no longer serve the role of inhibiting discrimination, at which point disclosure may contribute to positive outcomes such as increased social connection, support, and well-being (Pachankis 2007; Ragins 2008).

The use of cross-lagged panel models allowed us to examine the first exploratory aim of examining reciprocal relations among minority stressors. Time 1 expectations of rejection was significantly and positively associated with Time 2 discrimination. Expectations of rejection was the only proximal variable to be significantly associated with discrimination at Time 2, which suggests that expectations of rejection may precede discrimination. This finding aligns with the idea that contextual, personal, and structural factors likely interact to influence perceptions of discrimination (Major and O'Brien 2005). Both the Minority Stress Model (Meyer 2003) and Psyhcological Mediation Framework (Hatzenbuehler 2009) acknowledge the role of structural forces. When LGB people are expecting rejection based on their sexual orientation, likely in situations where contextual or structural factors have created unwelcome environments (e.g., states that still have discriminatory laws based on sexual orientation), they may be more likely to experience discriminatory events.

Time 1 expectations of rejection was also significantly associated with greater Time 2 concealment, which may serve a protective function (Collins and Zimmerman 1983). When expectations of rejection are high concealment of one's LGB status may be used to prevent potential negative consequences. Also regarding concealment, Time 1 concealment was positively associated with Time 2 internalized homophobia (Bruce et al. 2015). Concealment may have the potential to strengthen internal homophobic attitudes. The notion that concealment of a stigmatized identity contributes to negative views of self has been found across many stigmatized groups such as racial and ethnic minorities, the unemployed, those with mental illnesses, and sexual minorities (Pachankis 2007).

The autogregressive paths in our models allowed us to examine the second exploratory aim of assessing the stability of our measures over 1 month. All the paths evidenced a great deal of stability. This finding suggests that, among our sample, the experience of both distal and proximal stressors was consistent over 1 month. The stability of these paths may have been the result of the time frame of the study. Greater variability may be present in studies using shorter or longer assessment windows.

Practical Implications

Working collaboratively with LGB clients may help both clients and practitioners to understand the sociocultural contexts that contribute to the development and experience of minority stressors. Identifying structural elements that may be influencing expectations of rejection may help to promote acceptance and empowerment. This approach mirrors multicultural-feminist therapies, which move beyond the individual and toward consideration of influential external forces. Because structural factors have the potential to influence minority stressors, we recommend that practitioners take on advocacy

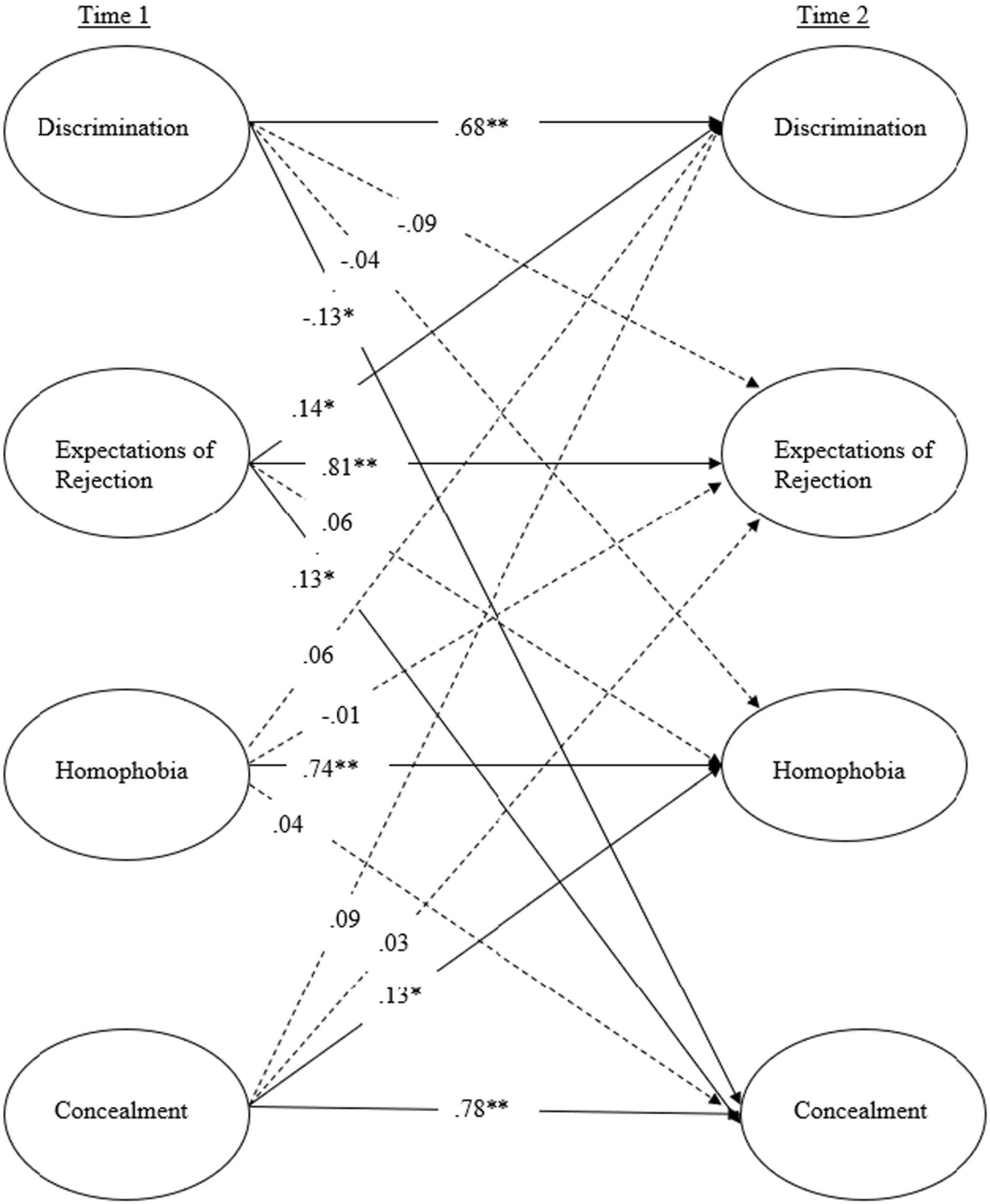

Fig. 1 Final structural model. Homophobia = internalized homophobia and concealment = sexual orientation identity concealment. *$p < .05$, **$p < .001$

roles in support of their sexual minority clients, which is in line with APA competencies (APA 2012). Our hopes are that these efforts help to reduce expectations of rejection and its consequences for LGB clients.

Table 3 Factor loadings for the final structural model

	Factor loading
Discrimination W1 P1	.958
Discrimination W1 P2	.886
Discrimination W1 P3	.895
Expectations W1 P1	.816
Expectations W1 P2	.880
Expectations W1 P3	.681
Concealment W1 P1	.866
Concealment W1 P2	.897
Concealment W1 P3	.944
Homophobia W1 Q1	.887
Homophobia W1 Q2	.652
Homophobia W1 Q3	.845
Homophobia W1 Q4	.832
Homophobia W1 Q5	.738
Discrimination W2 P1	.949
Discrimination W2 P2	.926
Discrimination W2 P3	.899
Expectations W2 P1	.844
Expectations W2 P2	.931
Expectations W2 P3	.694
Concealment W2 P1	.881
Concealment W2 P2	.932
Concealment W2 P3	.927
Homophobia W2 Q1	.880
Homophobia W2 Q2	.651
Homophobia W2 Q3	.873
Homophobia W2 Q4	.839
Homophobia W2 Q5	.706

Expectations expectations of rejection, *Homophobia* internalized homophobia, *Conceal* sexual orientation identity concealment, *W* Wave, *P* Parcel, and *Q* Question

Table 4 Final structural model covariances

Variable	1	2	3	4	5	6	7	8
1. Discrimination T1	–							
2. Expectations T1	.51**	–						
3. Homophobia T1	.22*	.06	–					
4. Conceal T1	.02	.11	.56**	–				
5. Discrimination T2					–			
6. Expectations T2					.31*	–		
7. Homophobia T2					.08	.17	–	
8. Conceal T2					.09	.04	.58**	–

Expectations expectations of rejection, *Homophobia* internalized homophobia, and *Conceal* sexual orientation identity concealment. Significant at $*p < .01$ and $**p < .001$ levels

We found that concealment predicted future internalized homophobia, or negative self-directed attitudes regarding one's sexual orientation. Practitioners may wish to explore reasons contributing to concealment and discuss both potential benefits (e.g., serving as a protective function; Collins and Zimmerman 1983) and consequences that can result from sexual orientation concealment (e.g., increased internalized homophobia). These conversations need to be considerate of each client's unique social and cultural context. It may be helpful to also discuss the potential benefits and consequences of identity disclosure (e.g., increased social connection and support; Collins and Zimmerman 1983; Ragins 2008) and the idea that disclosure occurs along a continuum (Ragins 2008). Concealment and disclosure may be dynamic, situation-dependent processes with the potential for both positive and negative consequences. A nuanced clinical approach that acknowledges these complexities and avoids assumptions of expertise on the part of the clinician would be most effective.

Limitations and Future Directions

A major strength of this study is the examination of the relations between distal and proximal stressors across two time points. Although this study alone cannot establish patterns of causality, our findings can help to guide future research. The limitations of this study provide directions for future research using minority stress models. First, because we only collected responses at two time points, we were unable to examine indirect effects (Cole and Maxwell 2003). Studies with at least three measurement points would allow for a more comprehensive examination of minority stress frameworks consisting of the distal and proximal stressors along with mental health outcomes. We did not examine physical or mental health outcomes because we first sought to examine how the relations between distal and proximal stressors manifested. More comprehensive studies of stressors and outcomes over more time points will also allow for an examination of non-linear trajectories, which may have been missed due to this study's design.

In addition to not including physical or mental health outcomes, this study did not assess any sources of resilience. Future research should include potential stress-buffering factors (e.g., social support, community connectedness), which are included in minority stress models (e.g., Hatzenbuehler 2009; Meyer 2003). Incorporating resilience factors will require thoughtful study design because scholars have suggested that studies conducted over longer spans, such as data collected 6 months apart, may allow participants more time to cope with discriminatory experiences (Herek et al. 1999; Swim et al. 2009).

Another limitation is that approximately 80% of our participants identified as White. This point is especially critical because past research has found that LGB people of color have varying relations among stigma, internalized homophobia, and outness when compared with White LGB individuals (Moradi et al. 2010). Researchers should also attempt to recruit and retain participants that may be experiencing greater levels of minority stress than typically reported because participants who dropped out of this study reported greater levels of discrimination and internalized homophobia. To inform work and interventions with LGB people experiencing varying levels of minority stress, it is necessary to recruit participants with a range of experiences. Similarly, future studies should incorporate the experiences of gender minorities. Although the Minority Stress Model (Meyer 2003) was originally intended to explain the experiences of sexual minorities, the model has since been expanded to address the unique experiences of people with varying gender identities (Hendricks and Testa 2012). Our sample consisted of some individuals of diverse gender identities, but our measures only captured the experiences of sexual versus gender minority stress.

Lastly, future research should attempt to replicate these findings using measures of heterosexist discrimination and sexual orientation concealment developed specifically for the experiences of sexual minorities. Our measure of discrimination was adapted from a measure developed to assess discrimination based on race/ethnicity, and our concealment measure originally assessed a "secret" that was not necessarily related to sexual orientation. These measures may not have fully captured the discrimination experiences of LGB people. Although the bivariate associations of our minority stressors were similar to past minority stress research it's possible the cross-lagged results may differ if other measures are used.

Conclusion

Our findings highlight that there may be bidirectional relations among distal and proximal stressors as outlined in the Psychological Mediation Framework (Hatzenbuehler 2009). Within minority stress frameworks variables in cross-sectional studies are often linearly positioned despite little empirical data to support such temporal connections. Until future research tests the replicability of our findings, and generalizability to other stressors, researchers should carefully consider the implied causal positioning of stressors described in the Minority Stress Model (Meyer 2003). The results of this study add to the growing call for minority stress frameworks to undergo longitudinal examination, which would allow scholars to speak more confidently about theorized temporal relations. We also hope these findings inform clinical work with LGB clients by underscoring the need to consider the nuanced relations that exist among minority stressors instead of assuming that some experiences necessarily contribute to others.

Compliance with Ethical Standards

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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